TUTOR-60

Matthew Reda · axiom

Author Note

In a trial randomizing 120 schools to different dose allocations, the analysis addresses whether buying fewer tutoring minutes for every eligible student yields higher gains than the pilot allocation of ninety weekly minutes. With the effect of tutoring on gain identified via the backdoor route, a spline response model evaluated over 4000 posterior draws shows that the recommended allocation costing 745 USD per student reaches 12000 students, whereas the pilot allocation reaches 4878 students. Relative to the pilot allocation, the recommended allocation delivers an estimated 13838 cohort reading points per year over the pilot allocation (90% HDI 6163 to 22036 points/year) against a decision threshold of 0 points/year. The posterior probability that the recommendation beats the pilot is 0.997. At the department's valuation, the value of the difference is 19372837 USD (90% HDI 8627873 to 30850017 USD).

Abstract

The study examined whether buying fewer tutoring minutes for every eligible student beats buying the pilot allocation for the fraction of them the budget reaches. The question is answered by cohort reading points over the pilot allocation, against a decision threshold of 0 points/year. The effect is identified via the backdoor route as described in the methods. A randomized trial of 120 schools provided the panel data for fitting spline responses through PyMC. The comparison yielded cohort reading points over the pilot allocation of 13838 points/year (90% HDI 6163 to 22036 points/year). The value of the difference at the department's own valuation settled above the threshold, on the favourable side. The strongest standing assumption is that a reading point has the value the state assigns it, which is asserted. Spreading a fixed budget across more students by reducing per-student intensity delivers larger total cohort gains than concentrating resources at the pilot level.

Reported quantities: Cohort reading points over the pilot allocation; Value of the difference at the department's own valuation

Introduction

This report addresses whether buying fewer tutoring minutes for every eligible student beats buying the pilot's ninety weekly minutes for the fraction of them the budget reaches. The question is answered by cohort reading points over the pilot allocation. This quantity is identified. The decision turns against a threshold of 0 points/year. An interval estimate lying wholly on one side of 0 points/year settles the question. An interval that spans 0 points/year leaves the question unsettled. A spanning interval does not indicate an absence of an effect.

Methods

To evaluate the question identified in the introduction, the analysis targets the effect of tutoring on gain, which is identified via the backdoor route. The identification graph includes arrows from poverty share to gain, resources to gain, resources to poverty share, and tutoring to gain, where resources are unmeasured, as shown in Figure 1. With 9,000,000 USD for one school year across 12,000 students reading two or more grade levels behind, or 750 USD per student if all are reached, both treatments are dosed in dollars per eligible student per year, setting 20 USD per weekly tutoring minute and 9 USD per weekly caregiver message up to a cap of 5 messages. Dosing tutoring in minutes and messaging in messages was considered and rejected because the two cannot then be traded against one another under the budget constraint. The trial randomized 120 schools to allocations drawn from a lattice over the two doses, with 3 messaging levels and tutoring spread across the decision range, as recorded in Table 1. Randomizing schools removes the arrows from poverty share to tutoring and resources to tutoring that downgrade observational records from 240 schools, while spreading allocations captures the shape of the response curve rather than conducting a two-arm trial at 90 minutes. A penalized spline response in each dose was fitted to the school-level panel through PyMC across 4,000 posterior draws. Within each draw, the per-student gain was multiplied by the number of students served at that allocation price to calculate total cohort reading points. Reporting the per-student response alone was rejected because multiplying post hoc drops the covariance across posterior draws and misstates the interval. The allocation that maximizes expected cohort points is compared with funding the pilot's 90 weekly minutes for the 4,878 students reached by that budget, pairing the comparison inside each draw to report the distribution of the difference rather than comparing against a baseline of no program. The procedure relies on two unresolved assumptions, named that a reading point has the value the state assigns it and that the cohort average answers the question.

Planning the trial

```
assignment: (tutoring_s, messaging_s) ~ Uniform(lattice) per
school s
observational graph : poverty_share -> gain, poverty_share ->
tutoring, resources -> gain, resources -> poverty_share,
resources -> tutoring, tutoring -> gain
status downgraded - requires the unmeasured node
randomized graph : poverty_share -> gain, resources -> gain,
resources -> poverty_share, tutoring -> gain
status identified via backdoor
```

Estimating the response

```
gain_s = f(tutoring_s) + g(messaging_s) + eps_s
f, g = penalised splines, one per dose, on the school-level
panel
effect(a) = E[gain | dose = a] - E[gain | dose = 0] within each
draw
served(a) = min(N_eligible, Budget / cost(a))
cohort(a) = served(a) * effect(a) the objective
cost(a) = a_tutoring + a_messaging (USD per eligible
student-year)
```

Choosing the allocation

```
a* = argmax_a mean_draws cohort(a)
Delta = cohort(a*) - cohort(a_pilot) paired inside each draw
P = Pr(Delta > 0)
value = Delta * VALUE_PER_POINT
```

The identification graph

From	Arrow	To	Kind
poverty_share	→	gain	direct effect
resources	→	gain	direct effect
resources	→	poverty_share	direct effect
tutoring	→	gain	direct effect
resources			unmeasured node

Table 1. The identification graph, as arrows

Assumptions this analysis rests on

Assumption	Statement	Status	Challenged by
a reading point has the value the state assigns it	a reading point is valued at 1,400 USD; the recommendation does not depend on this number, only on whether to fund at all	was asserted without a check	a valuation low enough to make the whole program not worth funding

the cohort average answers the question	the recommendation is made on the cohort average; the grade-band breakdown the program office asked for is not estimable from a school-randomized trial	is assumed and has not been checked	a response that differs across grade bands enough to change the allocation

Table 2. Standing assumptions

identification Schools were randomized to allocations; the adjustment set is empty.

assumes *schools_were_randomized_to_allocations* (satisfied): schools were randomized to allocations, so the adjustment set is empty

estimand The recommendation is over assigned allocations. Schools delivered 93% of assigned minutes, and that shortfall is inside the estimate rather than corrected out of it.

assumes *the_estimand_is_over_assigned_allocations* (satisfied): the recommendation is over assigned allocations; schools delivered 93% of assigned minutes and that shortfall is priced into the estimate

interval Every figure and quantity at 90% over 4,000 posterior draws.

limitation The grade-band breakdown the program office asked for is not estimable from a school-randomized trial, and is not reported.

assumes *the_cohort_average_answers_the_question* (unverified): the recommendation is made on the cohort average; the grade-band breakdown the program office asked for is not estimable from a school-randomized trial

Assumption ledger

Results

The analysis evaluates TUTOR-60 relative to the pilot allocation using the design and identification assumptions established in the methods. Table 2 reports the estimated quantities with their intervals and thresholds. Cohort reading points over the pilot allocation are 13838 points/year (90% HDI 6163 to 22036 points/year). This outcome measures performance against the pilot allocation rather than an absence of tutoring. The value of the difference at the department's own valuation is 19372837 USD (90% HDI 8627873 to 30850017 USD). This economic measure reflects the decision value introduced earlier in the report. The estimated quantities with their intervals are shown in Figure 2. The dashed rule in Figure 2 marks the decision threshold, where any interval crossing the line represents an unsettled result rather than a null effect.

COHORT READING POINTS OVER THE PILOT ALLOCATION	VALUE OF THE DIFFERENCE AT THE DEPARTMENT'S OWN VALUATION
13,838 points/year	19,372,837 USD
[6,200, 22,000] (90% HDI)	[9,000,000, 3.1e+07] (90% HDI)

What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

The recommendation is flat between 700 and 750 USD — 35 to 38 weekly minutes are all within two per cent of the best. The decision that matters is not the exact figure, it is not funding 90 minutes.

For the pilot to be the better buy, the advantage would have to be overstated by 100% of itself — that is, essentially the whole of it would have to be an artefact of the fit rather than a real difference between the two allocations.

The reason the recommendation is not 90 minutes is not that 90 minutes fails. It is that the response is steep at the bottom, the budget is fixed, and every extra dollar a student is a student not served.

Model checking

Model diagnostics describe the computational performance and sample properties of the fitted procedure. A check of this kind can lower confidence in an estimate, but it can never establish one. The estimation was conducted using 4000 posterior draws. Across the experimental design, 120 schools were randomized. The resulting posterior probability that the recommendation beats the pilot is 0.997. Under the recommended policy, the intervention reaches 12000 students, whereas the pilot allocation would reach 4878 students. The cost of the recommended allocation is 745 USD per student. Table 3 lists these recorded diagnostics. These metrics examine numerical behavior and policy coverage within the sample, leaving the broader identification assumptions from the methods unexamined.

POSTERIOR PROBABILITY THE RECOMMENDATION BEATS THE PILOT	STUDENTS THE RECOMMENDATION REACHES	STUDENTS THE PILOT ALLOCATION WOULD REACH	COST OF THE RECOMMENDED ALLOCATION PER STUDENT	SCHOOLS RANDOMIZED	POSTERIOR DRAWS
0.997	12,000	4,878	745 USD	120	4,000

Discussion

Both the value of the difference at the department's own valuation and cohort reading points over the pilot allocation settled above the threshold, on the favourable side. Because the findings agree, both measures indicate that distributing tutoring minutes across all eligible students improves outcomes relative to the pilot allocation. Because the analysis is identified via the backdoor route detailed in the methods, these estimates license an interpretation of policy effects rather than mere associations. This interpretation holds under the structure of that design and extends no further. The findings remain conditional on 2 unresolved assumptions. What each one claims is stated in the limitations section.

Conclusions

Buying fewer tutoring minutes for every eligible student beats buying the pilot's ninety weekly minutes for the fraction of students the budget reaches. The cohort reading points over the pilot allocation is 13838 points/year (90% HDI 6163 to 22036 points/year), which is settled above the threshold on the favourable side. The value of the difference at the department's own valuation is also settled above the threshold on the favourable side. Because the effect is identified, these findings support shifting resources to cover all eligible students. This conclusion remains conditional on 2 unresolved assumptions: that a reading point has the value the state assigns it, and that the cohort average answers the question. What each assumption claims is stated in the limitations section.

Limitations

Two unresolved assumptions carry aspects of the analysis that the data does not settle. First, the analysis asserts that a reading point has the value the state assigns it, valuing a reading point at 1,400 USD. This assumption licenses the recommendation to turn not on this specific dollar figure, but only on whether to fund the program at all. That stance would be challenged by a valuation low enough to make the whole program not worth funding. Second, the analysis relies on the unverified assumption that the cohort average answers the question. This licenses making the recommendation on the cohort average, because the grade-band breakdown requested by the program office is not estimable from a school-randomized trial. This second assumption would be challenged by a response that differs across grade bands enough to change the allocation.

Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
Cohort reading points over the pilot allocation	advantage	13838 points/year (90% HDI 6163 to 22036 points/year)	paired posterior comparison over the allocation grid
Value of the difference at the department's own valuation	advantage_value	19372837 USD (90% HDI 8627873 to 30850017 USD)	the same draws, priced
Posterior probability the recommendation beats the pilot	probability_better	0.997	—
Students the recommendation reaches	students_reached	12000	—
Students the pilot allocation would reach	pilot_reached	4878	—
Cost of the recommended allocation per student	cost_per_student	745 USD	—
Schools randomized	schools	120	—
Posterior draws	draws	4000	—

Table 3. Quantities and their sources

Field	Value
estimator	spline response surface, PyMC via nutpie
fit_hash	c878161a12a5
interval_mass	90%
seed	0
world	nbs/case-studies/tutoring/tutoring.py
evidence hash	7bc9d4b50db1c7e7d179ec9a8af11003732629bc0189790cf210b75d82f12a52

narration: abstract	gemini-3.5-flash-lite — verified
narration: title_page	gemini-3.7-flash — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified
narration: diagnostics	gemini-3.7-flash — verified
narration: discussion	gemini-3.7-flash — verified
narration: conclusions	gemini-3.7-flash — verified
narration: limitations	gemini-3.7-flash — verified

Table 4. Run

Tables

Table 1. The design, as recorded parameters.

Step	Parameter	Value
Identification	graph	poverty_share -> gain, resources -> gain, resources -> poverty_share, tutoring -> gain
Identification	graph hash	488a01d83085
Identification	route	backdoor
Identification	status	identified
Identification	unmeasured nodes	resources
The decision as arithmetic	budget (USD)	9,000,000
The decision as arithmetic	eligible students	12,000
The decision as arithmetic	messaging dose range (USD)	0 to 45
The decision as arithmetic	per student if all are reached (USD)	750
The decision as arithmetic	tutoring dose range (USD)	0 to 3,000
Planning the trial	eligible per school	35–65
Planning the trial	messaging levels	0, 18, 45
Planning the trial	schools	120
Planning the trial	unit of assignment	school
Estimating the response	fit hash	c878161a12a5
Estimating the response	interval mass	90%
Estimating the response	posterior draws	4,000
Estimating the response	response family	spline in both doses
Estimating the response	sampler	PyMC via nutpie
Choosing the allocation	allocate() at the budget	tutoring 705, messaging 45 (converged)
Choosing the allocation	messages a week	5

Choosing the allocation	recommended messaging (USD)	45
Choosing the allocation	recommended tutoring (USD)	700
Choosing the allocation	students reached	12,000
Choosing the allocation	weekly minutes	35
Choosing the allocation	within 2% of the best (USD)	700 to 750

Table 2. Estimated quantities with their intervals and thresholds.

Quantity	Estimate	Unit	Interval	Threshold	Relative to threshold
Cohort reading points over the pilot allocation	13838	points/year	6163 to 22036 (90% HDI)	0	above threshold
Value of the difference at the department's own valuation	19372837	USD	8627873 to 30850017 (90% HDI)	0	above threshold

Table 3. Recorded diagnostics.

Check	Value	Unit	Interval
Posterior probability the recommendation beats the pilot	0.997	—	—
Students the recommendation reaches	12000	—	—
Students the pilot allocation would reach	4878	—	—
Cost of the recommended allocation per student	745	USD	—
Schools randomized	120	—	—
Posterior draws	4000	—	—

Figures

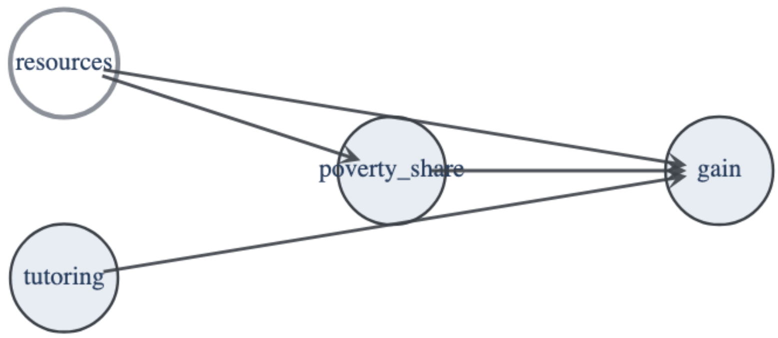

Figure 1. The identification graph. A hollow node was not measured and a dashed edge is an unmeasured common cause.

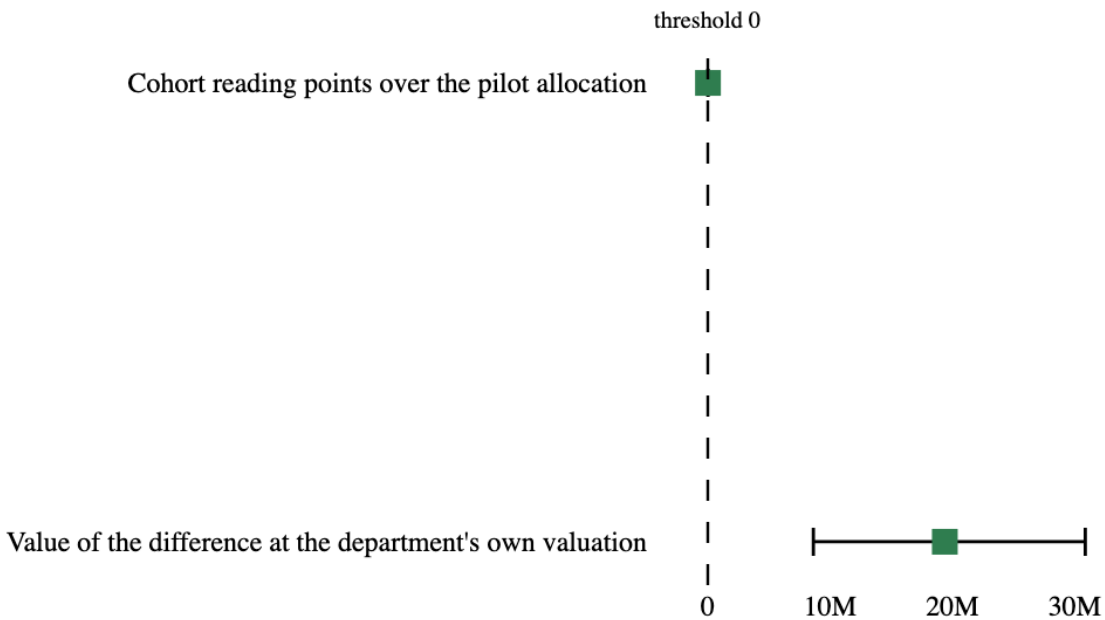

Figure 2. Estimated quantities with their intervals. The dashed rule marks the decision threshold; a row whose interval crosses it is unsettled rather than null.